

Astronomical Refraction



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DECLARATION FROM THE STUDENT

I affirm that I have identified all my sources and that no part of my dissertation paper uses unacknowledged materials.

Acknowledgement

The project entitled **ASTRONOMICAL REFRACTION** comprises of my research work. I would like to express my sincere gratitude towards my supervisor, Professor **Diptiman Saha**, Associate Professor, Department of Mathematics, SXC Kolkata; without whom this project would not have been possible. His constant support and encouragement, motivated me to successfully complete my dissertation.

I would also like to thank Professor **Sucharita Roy**, H.O.D, Assistant Professor, Department of Mathematics and St. Xavier's College (Autonomous), Kolkata, for providing me with the opportunity to undertake this topic as my project.

Lastly, I would like to thank my parents Mr. Prabal Sen , Mrs. Jhumur Sen and my friends for being supportive throughout the process of this project.

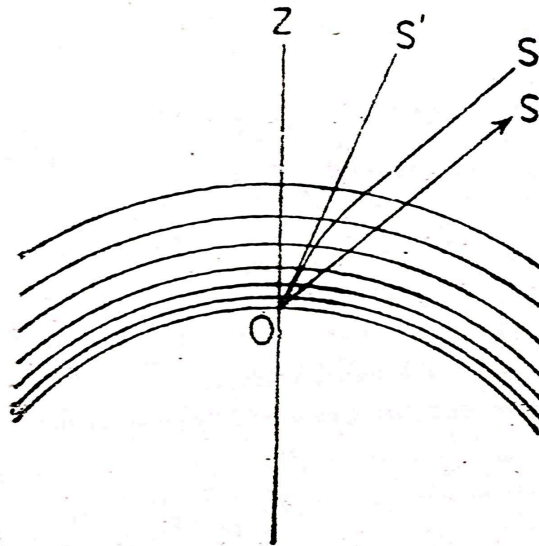
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1 Introduction

A ray of light in passing from one medium to another undergoes a change in its direction at the surface of separation of the two mediums. This change of direction is called refraction. If the ray enters from a rarer medium to a denser medium the deviation is towards the normal of the surface of separation at the point where the ray meets the surface.

Consider a Ray of light proceeding from a star to an observer. As it passes through different layers of atmosphere (where the density increases from layer to layer as we go downwards) the rays bend towards the normal at the surface of separation of each layer. The actual direction of the star is OS but the apparent direction is OS' . (The ray appears to reach the observer from the direction $S'O$). Thus on account of refraction it appears that the star is raised in the heavens in other words its Zenith distance decreases. But as the incident ray and the refracted ray lie in the same vertical plane the azimuth remains unaffected.



2 Historical background

Astronomical refraction has had a long and fascinating history. Cleomedes (100 A.D.) and Ptolemy (200 A.D.) were aware of its existence and understood in a qualitative way some of its properties. Alhazen (1100 A.D.) quite correctly suggested that the flattening of the sun's disk near the horizon was due to astronomical refraction. Tycho Brahe in 1587, however, was the first to make direct measurements of the magnitude of the refraction. The first theory of astronomical refraction based on Snell's law was that of Cassini, who in 1656 looked upon the earth's atmosphere as being of constant refractive index up to its upper limit at which all the refraction took place. Extensive investigations of the physical properties of the atmosphere modified these ideas and led to the "concentric spherical shell model" and the "plane parallel layer model," the latter being an approximation for the former. Attempts to evaluate the "refraction integral" for the concentric spherical shell model have led to the theories of Bessel, Bradley, Gylden, Ivory, Laplace, Mayer, Simpson, Young, and others. At the beginning of the twentieth century, investigations on the physical properties of the atmosphere extended to higher elevations, and new physical properties appeared which were not anticipated by the earlier workers. This led to the celebrated work of Harzer, who for the first time was able to compute the astronomical refraction purely from meteorological measurements. Harzer also made the first detailed investigation of the applicability of the spherical shell model, which is used almost universally by workers in this field. Harzer's work has served as a source of inspiration, but much remains to be done. This paper attempts to give a brief chronological description of some of the more significant theories of astronomical refraction for the purpose of pointing out the types of problems which have existed, how these problems were attacked, and the problems which still remain. It also gives adequate references for those wishing additional information.

3 The Formula for Astronomical Refraction

To show that for moderate zenith distance the refraction of a heavenly body (temperature and pressure of atmosphere remaining constant) varies as the tangent of the apparent or observed Zenith distance. The height of the atmosphere being small in comparison with the earth's radius we can assume for moderate values of the zenith the distance that the atmosphere consists of a large amount of horizontal layers in which the density increases as we go downwards. Again the refraction due to a number of mediums whose surfaces of separation are parallel planes is the same as that due to a single medium of density equal to that in the last medium.

Hence in finding the refraction of a Ray of light from a heavenly body, we may suppose the ray to be passing through a homogeneous atmosphere of the same density as that of the layer in contact with the Earth's surface regarded as a horizontal plane.

Let SPO be the path of a ray from a star to an observer at O on the surface of the earth so that OPS' is the apparent direction of the star. Let AB and CD be the surfaces of the Earth and the highest layer of the atmosphere respectively.

Now $\angle ZOS' =$ apparent zenith distance (z say), $\angle SPS' =$ amount of refraction (x say), so that the angle of incidence $\angle SPZ = z + x$ and angle of refraction $\angle OPN = \angle POZ = z$

$$\frac{\sin(z + x)}{\sin z} = \mu(\text{constant})$$

or, $\sin z \cos x + \cos z \sin x = \mu \sin z$

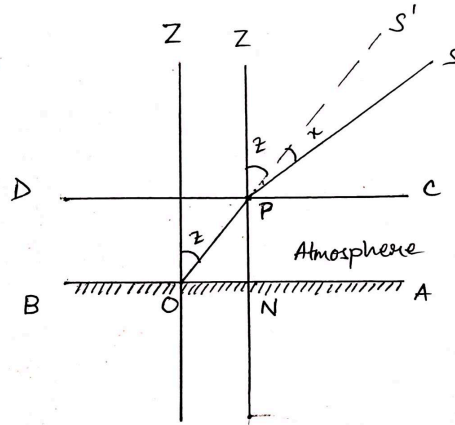
Since x is small, in circular measure, $\sin x = x$ and $\cos x = 1$, very nearly.

or, $x \cos z = (\mu - 1) \sin z$

$x = (\mu - 1) \tan z = k \tan z$, where $k = \mu - 1$

Thus *the amount of refraction varies as the tangent of the zenith distance.*

The constant k is called *the co-efficient of refraction.*



Observation

If z is observed zenith distance, true zenith distance $= z + k \tan z$

If α is observed altitude, true altitude $= \alpha - k \cot \alpha$

- Note.** (i) When $z = 90^\circ$, the tangent formula gives an absurd result for the amount of refraction x , in view of the fact that the value of tangent tends to infinity as x tends to 90°
- (ii) When $z = 0^\circ$, i.e., the body is at the zenith, $x = 0$; the position of the body is not affected by refraction.
- (iii) When $z = 45^\circ$, $x = k \tan 45^\circ = k$.

4 To find co-efficient of refraction

4.1 First Method

By observing the meridian zenith distances of a circumpolar star when (i) the latitude of the place is known, (ii) the declination of the star is known.

Let m, n be the real positions and m', n' the observed positions of a circumpolar star when crossing meridian above and below the pole respectively. Let z_1 and z_2 be the observed zenith distances at the two positions.

Now

$$z_1 + k \tan z_1 = Zm = Zp - Pm \quad (1)$$

$$z_2 + k \tan z_2 = Zn = ZP + Pn \quad (2)$$

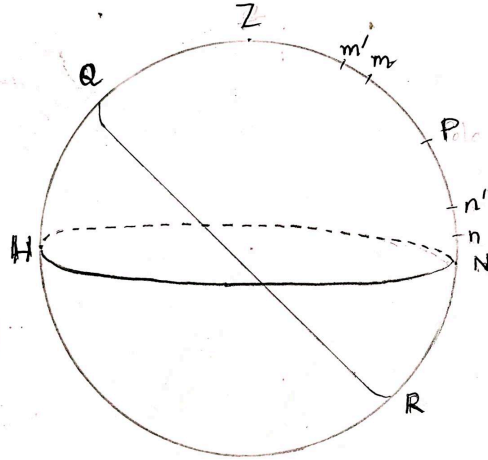
where $Pm = Pn$

Adding (1) and (2) we get

$$z_1 + z_2 + k(\tan z_1 + \tan z_2) = 2ZP = 2(90^\circ - \phi) \quad (3)$$

therefore,

$$k = \frac{180^\circ - 2\phi - z_1 - z_2}{\tan z_1 + \tan z_2}$$



Note. (i) A slight error in the value of ϕ affects the value of k

(ii) When the declination of the star is known, we subtract (2) from (1) and get $z_2 - z_1 + k(\tan z_2 - \tan z_1) = 2Pn = 2(90^\circ - \delta)$

therefore,

$$k = \frac{180^\circ - 2\delta - (z_2 - z_1)}{\tan z_1 - \tan z_2}$$

4.2 Second Method

By observing the meridian zenith distances (ζ_1, ζ_2) of a second circumpolar star, we get as in the case of the first circumpolar star.

$$\zeta_1 + \zeta_2 + k(\tan \zeta_1 + \tan \zeta_2) = 2(90^\circ - \phi) \quad (4)$$

Equating the values of (3) and (4), we have

$$z_1 + z_2 + k(\tan z_1 + \tan z_2) = \zeta_1 + \zeta_2 + k(\tan \zeta_1 + \tan \zeta_2)$$

From this result we obtain

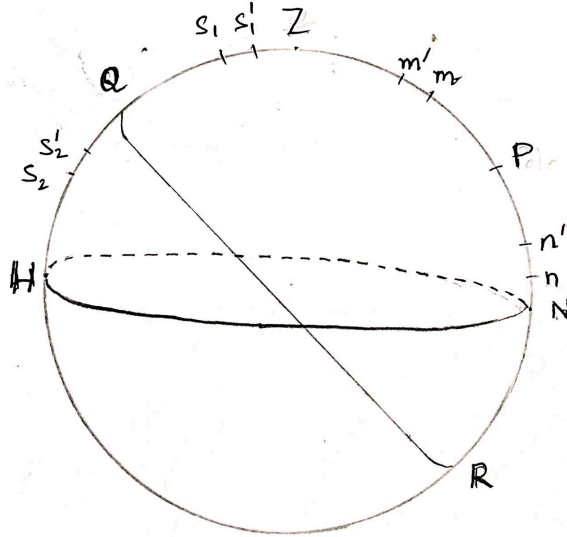
$$k = \frac{\zeta_1 + \zeta_2 - z_1 - z_2}{\tan z_1 + \tan z_2 - \tan \zeta_1 - \tan \zeta_2}$$

4.3 Bradley's Method

In Bradley's method the meridian zenith distances of a circumpolar star are observed. Also the meridian zenith distances of the sun at the two solstices are observed.

Let m, n be the real positions of and m', n' the observed positions of the star at the upper and lower transits: z_1, z_2 being the observed zenith distances respectively.

Let S_1, S_2 be the real positions of and S'_1, S'_2 the observed positions of the sun when crossing the meridian at the summer and winter solstices. Let s_1, s_2 be the observed zenith distances at the two positions.



We have,

$$z_1 + k \tan z_1 = Zm = ZP - Pm \quad (5)$$

$$z_2 + k \tan z_2 = Zn = ZP + Pn \quad (6)$$

therefore,

$$z_1 + z_2 + k(\tan z_1 + \tan z_2) = 2ZP = 2(90^\circ - \phi) \quad (7)$$

Again

$$s_1 + k \tan s_1 = ZS_1 = ZQ - QS_1 \quad (8)$$

$$s_2 + k \tan s_2 = ZS_2 = ZQ + QS_2 \quad (9)$$

where $QS_1 = QS_2 = \omega$ therefore,

$$s_1 + s_2 + k(\tan s_1 + \tan s_2) = 2ZQ = 2\phi \quad (10)$$

Adding (7) and (10)

$$z_1 + z_2 + s_1 + s_2 + k(\tan z_1 + \tan z_2 + \tan s_1 + \tan s_2) = 180^\circ$$

therefore,

$$k = \frac{180^\circ - z_1 - z_2 - s_1 - s_2}{\tan z_1 + \tan z_2 + \tan s_1 + \tan s_2}$$

Advantages and disadvantages. In Bradley's method, the advantage is that the latitude of the place need not be known. Any error due to slight error in the latitude will not affect the determination of the co-efficient of refraction. But the main disadvantage is that one has to wait six months for observing the two zenith distances of the sun.

5 Cassini's formula for atmospheric refraction

Let $SO'O$ be the path of a ray of light from a star S . Cassini assumes that this ray undergoes a single refraction on entering the *homogeneous but spherical* atmosphere at O' . Let O be the position of the observer and c the centre of the Earth.

let the earth's radius $CO=a$, n = the ratio of the height of the homogeneous atmosphere to the radius of the earth.

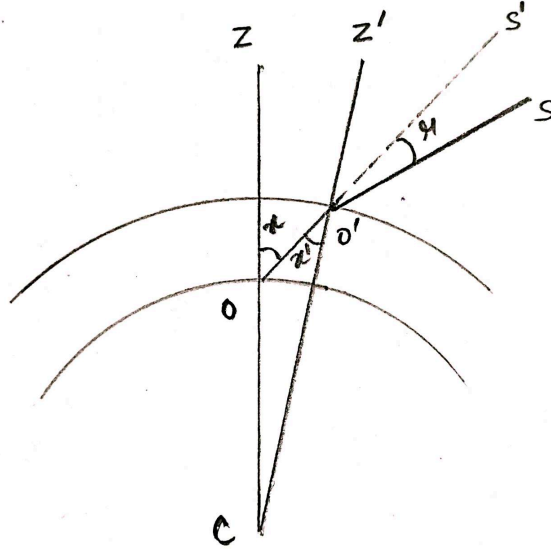
z = observed zenith distance, z' = the angle of refraction when the ray enters homogeneous atmosphere, r = amount of the refraction.

Now, the $\sin(z' + r) = \mu \sin z'$

or, $\sin z' \cos r + \cos z' \sin r = \mu \sin z'$

Since r is small, in circular measure, $\sin r = r$ and $\cos r = 1$, very nearly.

$\sin z' + \cos z' \cdot r = \mu \sin z'$; i.e., $r = (\mu - 1) \tan z'$



From the $\triangle OO'C$,

$$\frac{\sin z'}{\sin(180 - z)} = \frac{a}{a + na} \text{ or, } \sin z' = \frac{\sin z}{1 + n}$$

$$\begin{aligned} \tan z' &= \frac{\sin z}{\sqrt{(1 + n)^2 - \sin^2 z}} = \frac{\sin z}{\sqrt{\cos^2 z + 2n + n^2}} \\ &= \tan z(1 + (2n + n^2) \sec^2 z)^{-\frac{1}{2}} \\ &= \tan z(1 - n \sec^2 z) \end{aligned}$$

neglecting the square and the higher powers of n .

Now $r = (\mu - 1) \tan z(1 - n \sec^2 z)$

This formula is of the form $r = A \tan z + B \tan^3 z$

where $A = (\mu - 1)(1 - n)$, $B = -n(\mu - 1)$

$$= \frac{1}{15} \times \frac{R''}{\sqrt{\cos(\delta + \phi) \cos(\delta - \phi)}}$$

Also

$$\cos PS'N = \sin ZS'P = \frac{\sin(90^\circ - \phi) \sin(360^\circ - h)}{\sin 90^\circ} = -\sin h \cos \phi$$

[h being the hour angle when observed zenith distance = 90°]

Again substituting in first equation we get,

$$t = -\frac{1}{15} \times \frac{R''}{\sin h \cos \phi \cos \delta}$$

6.2 Second Method

Due to refraction a heavenly body is raised in the heavens, hence a body is seen earlier than when it is actually on the horizon.

We have,

$$\cos z = \sin \phi \sin \delta + \cos \phi \cos \delta \cos h$$

therefore,

$$-\sin z \Delta z = -\cos \phi \cos \delta \sin h \Delta h$$

or,

$$\Delta h = \frac{\sin z \Delta z}{\cos \phi \cos \delta \sin h}$$

Here $\sin z = \sin 90^\circ = 1$, $\Delta z = R''$, $\Delta h = -15t''$ (when rising)

Therefore,

$$-15t = \frac{R''}{\cos \phi \cos \delta \sin h}$$

or,

$$t = -\frac{1}{15} \times \frac{R''}{\sin h \cos \phi \cos \delta} \text{seconds}$$

When $z = 90^\circ$; $\cos h = -\tan \phi \tan \delta$, $\sin h = -\sqrt{1 - \tan^2 \phi \tan^2 \delta}$

Substituting we get,

$$t = \frac{1}{15} \times \frac{R''}{\sqrt{\cos(\delta + \phi) \cos(\phi - \delta)}}$$

Again from the spherical triangle ZPS;, by the sine formula,

$$-\sin c \cos \phi = \sin PS'Z = \cos PS'N$$

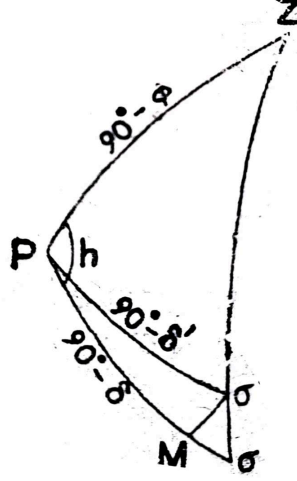
therefore,

$$\begin{aligned} t &= -\frac{1}{15} \times \frac{R''}{\sin h \cos \phi \cos \delta} \\ &= \frac{1}{15} \times \frac{R''}{\cos PS'N \cos \delta} \end{aligned}$$

7 Effect of refraction on H.A(or R.A) and declination of a star

Let σ be the real position of a star on the celestial sphere and σ' the displaced position along the vertical through σ .

Let the observed zenith distance $Z\sigma' = z$, then $\sigma\sigma' = k \tan z$. Let δ, δ' be the declination's and h, h' the hour angles of σ and σ' ; and $\angle P\sigma Z = \eta, \angle P\sigma' Z = \eta'$



Through σ' draw small circular arc whose pole is P to meet $P\sigma$ at M . Since $\sigma\sigma'$ is small we may regard $\sigma'M\sigma$ as a plane triangle.

(i) Change in declination

$$\begin{aligned}\delta - \delta' &= -\sigma M = -\sigma\sigma' \cos \eta \\ &= -k \tan z \cos \eta \\ &= -k \tan z \cos \eta'\end{aligned}$$

Also from spherical triangle $PZ\sigma'$

$$\begin{aligned}\sin z \cos \eta' &= \sin \phi \cos \delta' - \cos \phi \sin \delta' \cos h' \\ \delta - \delta' &= -k \sec z (\sin \phi \cos \delta' - \cos \phi \sin \delta' \cos h')\end{aligned}$$

(ii) Change in H.A, i.e.,

$$\begin{aligned}h - h' &= \angle \sigma P \sigma' = \frac{\sigma' M}{\sin P \sigma'} = \frac{\sigma \sigma' \sin \eta}{\cos \delta'} \\ &= k \tan z \sin \eta \sec \delta' = k \tan z \sin \eta' \sec \delta'\end{aligned}$$

Also from the spherical triangle $ZP\sigma$ by the sine rule

$$\sin z \sin \eta' = \sin(90^\circ - \phi) \sin h' = \cos \phi \sin h'$$

$$h - h' = k \cos \phi \sin h' \sec z \sec \delta'$$

δ', h' are derived from observations: therefore we get δ and h .

Note $\alpha - \alpha' = h' - h = -k \cos \phi \sin h' \sec z \sec \delta'$

8 Effect of refraction on a pair of neighbouring celestial bodies

Let S, S' be two bodies close to each other; d being the angular distance between them and z, z' their zenith distances. If Z is the zenith due to refraction they are raised along SZ and $S'Z$ to the positions S_1 and S'_1 .

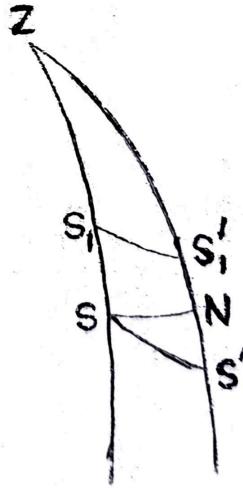
Let $\angle SZS' = \alpha$ and $\angle ZSS' = \theta$

Now $d^2 = (z' - z)^2 + \alpha^2 \sin^2 z$ (approx.)

therefore, $2d\Delta d = 2(z' - z)(\Delta z - \Delta z') + 2\alpha^2 \sin z \cos z \Delta z$

where $\Delta z' = -k \tan z', \Delta z = -k \tan z$

hence, $-d\Delta d = k(z' - z)(\tan z' - \tan z) + k\alpha^2 \sin^2 z$
 $= k \frac{(z' - z) \sin(z' - z)}{\cos z \cos z'} + k\alpha^2 \sin^2 z$



9 Oval shape of the sun and moon near the horizon

The sun and the moon appear to have oval shapes when near the horizon. The reason is that the lower limb being nearer the horizon than the upper limb will be raised to a greater amount. The vertical diameter AB appears diminished as A'B'. But both ends of the horizontal diameter having equal Zenith distances are raised through equal amounts so that the horizontal diameter remains the same. The apparent diminution of the vertical diameter is found to be about $\frac{1}{6}$ th of the whole.

10 Lengthening of Day time

Day begins when we just see the upper limb of the sun above the horizon and ends when the upper limb has just gone below the horizon. When sun's centre is seen at the horizon it is really 35 minutes below the horizon and when sun's upper limb is at the horizon sun's centre is 51 minutes below the horizon. In consequence the rising is accelerated and setting retarded so that the day time lengthens.

11 Effect of Refraction on Lunar Eclipse

In a total lunar eclipse the Earth coming between the sun and the moon prevents the sun's rays from reaching the moon directly. But those rays which nearly graze the Earth surface are bent by refraction of the earth's atmosphere and does reach the moon's disc. Also long red light waves have more penetrating power than short violet ones. On account of this, the moon's disc is never completely dark but appears of a dull red colour.

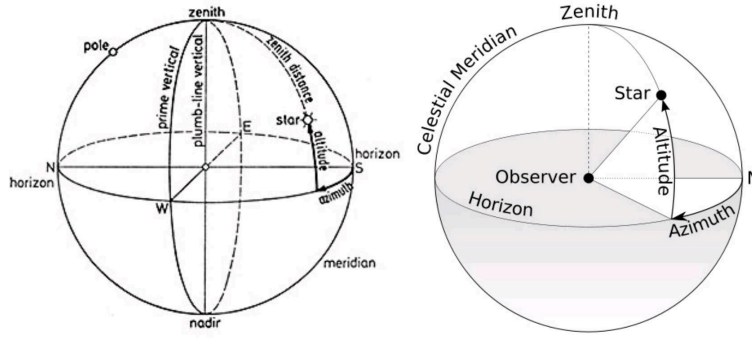


Figure 1: Celestial Sphere

12 Terminologies

The **Celestial sphere** is a sphere on which heavenly bodies appear to be depicted. The observer is at the centre of the celestial sphere and it is usual to regard it as a sphere of infinitely large radius compared with any distance on the earth, so that the Earth is a mere point at the centre of this sphere.

Zenith and Nadir. The point where the vertical line meets the celestial sphere overhead is called the Zenith. The point vertically below the celestial sphere is called the nadir.

Prime vertical is the great circle through the Zenith and the nadir and at right angles to the meridian. The prime vertical is the vertical through the east and the west point.

The **Azimuth** of a heavenly body is the arc of the horizon between the north point and the perpendicular Arc to the horizon drawn through the body it is measured from 0° to 180° towards east or west of the north point.

The **Altitude** of a heavenly body is its distance from the horizon (measured along the perpendicular Arc to the horizon drawn through the body). It is measured towards the Zenith from 0° to 90° .

The **Zenith** distance of a heavenly body is its distance from the Zenith measured along the secondary to the horizon drawn through the body.

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